

HOMER- PROFORMA

Hospital ID:	HOMER ID:
Date of assessment:	Age in years:
Sex: ☐ Male ☐ Female ☐ Others	Marital status: ☐ Single ☐ Married ☐ Others
Language: ☐ English ☐ Kannada ☐ Tamil ☐ Telugu ☐ Hindi ☐ Punjabi	Education level: ☐ No formal education ☐ Primary (1-5) ☐ Secondary (6-10) ☐ Intermediate (PUC/Diploma) ☐ Graduate ☐ Post-graduate/ Honors
Hand dominance: ☐ Right ☐ Left	Occupational status: ☐ Employed ☐ Unemployed ☐ Retired
Co-morbidities:	
Premorbid walking status: ☐ Independent with/without gait aid ☐ With assistance ☐ Unable to commute	Active hand movements at stroke onset? ☐ Yes ☐ No
Stroke severity: NIHSS scores	Time since stroke (in days):
Stroke type: ☐ Ischemic ☐ Haemorrhagic ☐ Ischemia with haemorrhagic transformation	Stroke location: ☐ Cortical ☐ Subcortical
Hemiparetic/hemiplegic side: ☐ Right ☐ Left	Thrombolysis/reperfusion therapy: ☐ Yes ☐ No
Diagnosis confirmed by: ☐ CT ☐ MRI	Aphasia: ☐ Present ☐ Absent

Outcome measures	A0	A1	A2
Fugl-Meyer assessment- UE total scores			
Action Research Arm Test total scores			
Motor Activity Log			
• Mean amount score			
• Mean how well score			
Chedoke Arm and Hand Activity Inventory total scores			
Robotic Assessments- PLUTO			
• Wrist flexion			
• Wrist extension			
• Wrist ulnar deviation			
• Wrist radial deviation			
• Forearm pronation			
• Forearm supination			
• Hand opening and closing			
Robotic Assessments- MARS			
• Medial and lateral movement of shoulder			
• Anterior and posterior movement of elbow			
• Combination (MLAP)			
EuroQol-5D-5L			
Subjective Index of Physical and Social Outcome total scores			
Modified Rankin Scale scores			
Caregiver Strain Index scores			
Modified Ashworth Scale			
• Shoulder flexors			
• Shoulder extensors			
• Shoulder abductors			
• Shoulder adductors			
• Elbow flexors			
• Elbow extensors			
• Forearm pronators			
• Forearm supinators			
• Wrist flexors			
• Wrist extensors			
Box and Block test scores			
Fatigue Severity Scale scores			
Patient Health Questionnaire-9 scores			
Actigraph			-

FUGL-MEYER ASSESSMENT **UPPER EXTREMITY (FMA-UE)** **Assessment of sensorimotor function**

ID:
Date:
Examiner:

Fugl-Meyer AR, Jaasko L, Leyman I, Olsson S, Steglind S: The post-stroke hemiplegic patient. A method for evaluation of physical performance. Scand J Rehabil Med 1975, 7:13-31.

A. UPPER EXTREMITY, sitting position					
I. Reflex activity		none	can be elicited		
Flexors: biceps and finger flexors (at least one)		0	2		
Extensors: triceps		0	2		
Subtotal I (max 4)					
II. Volitional movement within synergies, without gravitational help		none	partial	full	
Flexor synergy: Hand from contralateral knee to ipsilateral ear. From extensor synergy (shoulder adduction/ internal rotation, elbow extension, forearm pronation) to flexor synergy (shoulder abduction/ external rotation, elbow flexion, forearm supination). Extensor synergy: Hand from ipsilateral ear to the contralateral knee	Shoulder	retraction	0	1	2
		elevation	0	1	2
		abduction (90°)	0	1	2
		external rotation	0	1	2
	Elbow	flexion	0	1	2
	Forearm	supination	0	1	2
	Shoulder	adduction/internal rotation	0	1	2
	Elbow	extension	0	1	2
	Forearm	pronation	0	1	2
	Subtotal II (max 18)				
III. Volitional movement mixing synergies, without compensation		none	partial	full	
Hand to lumbar spine hand on lap	cannot perform or hand in front of ant-sup iliac spine hand behind ant-sup iliac spine (without compensation) hand to lumbar spine (without compensation)	0	1	2	
Shoulder flexion 0°- 90° elbow at 0° pronation-supination 0°	immediate abduction or elbow flexion abduction or elbow flexion during movement flexion 90°, no shoulder abduction or elbow flexion	0	1	2	
Pronation-supination elbow at 90° shoulder at 0°	no pronation/supination, starting position impossible limited pronation/supination, maintains starting position full pronation/supination, maintains starting position	0	1	2	
Subtotal III (max 6)					
IV. Volitional movement with little or no synergy		none	partial	full	
Shoulder abduction 0 - 90° elbow at 0° forearm pronated	immediate supination or elbow flexion supination or elbow flexion during movement abduction 90°, maintains extension and pronation	0	1	2	
Shoulder flexion 90° - 180° elbow at 0° pronation-supination 0°	immediate abduction or elbow flexion abduction or elbow flexion during movement flexion 180°, no shoulder abduction or elbow flexion	0	1	2	
Pronation/supination elbow at 0° shoulder at 30°- 90° flexion	no pronation/supination, starting position impossible limited pronation/supination, maintains start position full pronation/supination, maintains starting position	0	1	2	
Subtotal IV (max 6)					
V. Normal reflex activity assessed only if full score of 6 points is achieved in part IV; compare with the unaffected side		0 (IV), hyper	lively	normal	
biceps, triceps, finger flexors	2 of 3 reflexes markedly hyperactive or 0 points in part IV 1 reflex markedly hyperactive or at least 2 reflexes lively maximum of 1 reflex lively, none hyperactive	0	1	2	
Subtotal V (max 2)					
Total A (max 36)					

B. WRIST support may be provided at the elbow to take or hold the starting position, no support at wrist, check the passive range of motion prior testing		none	partial	full
Stability at 15° dorsiflexion elbow at 90°, forearm pronated shoulder at 0°	less than 15° active dorsiflexion dorsiflexion 15°, no resistance tolerated maintains dorsiflexion against resistance	0	1	2
Repeated dorsiflexion / volar flexion elbow at 90°, forearm pronated shoulder at 0°, slight finger flexion	cannot perform volitionally limited active range of motion full active range of motion, smoothly	0	1	2
Stability at 15° dorsiflexion elbow at 0°, forearm pronated slight shoulder flexion/abduction	less than 15° active dorsiflexion dorsiflexion 15°, no resistance tolerated maintains dorsiflexion against resistance	0	1	2
Repeated dorsiflexion / volar flexion elbow at 0°, forearm pronated slight shoulder flexion/abduction	cannot perform volitionally limited active range of motion full active range of motion, smoothly	0	1	2
Circumduction elbow at 90°, forearm pronated shoulder at 0°	cannot perform volitionally jerky movement or incomplete complete and smooth circumduction	0	1	2
Total B (max 10)				

C. HAND support may be provided at the elbow to keep 90° flexion, no support at the wrist, compare with unaffected hand, the objects are interposed, active grasp		none	partial	full
Mass flexion from full active or passive extension		0	1	2
Mass extension from full active or passive flexion		0	1	2
GRASP				
a. Hook grasp flexion in PIP and DIP (digits II-V), extension in MCP II-V	cannot be performed can hold position but weak maintains position against resistance	0	1	2
b. Thumb adduction 1-st CMC, MCP, IP at 0°, scrap of paper between thumb and 2-nd MCP joint	cannot be performed can hold paper but not against tug can hold paper against a tug	0	1	2
c. Pincer grasp, opposition pulpa of the thumb against the pulpa of 2-nd finger, pencil, tug upward	cannot be performed can hold pencil but not against tug can hold pencil against a tug	0	1	2
d. Cylinder grasp cylinder shaped object (small can) tug upward, opposition of thumb and fingers	cannot be performed can hold cylinder but not against tug can hold cylinder against a tug	0	1	2
e. Spherical grasp fingers in abduction/flexion, thumb opposed, tennis ball, tug away	cannot be performed can hold ball but not against tug can hold ball against a tug	0	1	2
Total C (max 14)				

D. COORDINATION/SPEED , sitting, after one trial with both arms, eyes closed, tip of the index finger from knee to nose, 5 times as fast as possible		marked	slight	none
Tremor	at least 1 completed movement	0	1	2
Dysmetria at least 1 completed movement	pronounced or unsystematic slight and systematic no dysmetria	0	1	2
		≥ 6s	2 - 5s	< 2s
Time start and end with the hand on the knee	at least 6 seconds slower than unaffected side 2-5 seconds slower than unaffected side less than 2 seconds difference	0	1	2
Total D (max 6)				

TOTAL A-D (max 66)				
H. SENSATION , upper extremity eyes closed, compared with the unaffected side		anesthesia	hypoesthesia or dysesthesia	normal
Light touch	upper arm, forearm	0	1	2
	palmary surface of the hand	0	1	2
		less than 3/4 correct or absence	3/4 correct or considerable difference	correct 100%, little or no difference
Position small alterations in the position	shoulder	0	1	2
	elbow	0	1	2
	wrist	0	1	2
	thumb (IP-joint)	0	1	2
Total H (max12)				

J. PASSIVE JOINT MOTION , upper extremity, sitting position, compare with the unaffected side				J. JOINT PAIN during passive motion, upper extremity		
	only few degrees (less than 10° in shoulder)	decreased	normal	pronounced pain during movement or very marked pain at the end of the movement	some pain	no pain
Shoulder						
Flexion (0° - 180°)	0	1	2	0	1	2
Abduction (0°-90°)	0	1	2	0	1	2
External rotation	0	1	2	0	1	2
Internal rotation	0	1	2	0	1	2
Elbow						
Flexion	0	1	2	0	1	2
Extension	0	1	2	0	1	2
Forearm						
Pronation	0	1	2	0	1	2
Supination	0	1	2	0	1	2
Wrist						
Flexion	0	1	2	0	1	2
Extension	0	1	2	0	1	2
Fingers						
Flexion	0	1	2	0	1	2
Extension	0	1	2	0	1	2
Total (max 24)				Total (max 24)		

A. UPPER EXTREMITY	/36
B. WRIST	/10
C. HAND	/14
D. COORDINATION / SPEED	/ 6
TOTAL A-D (motor function)	/66

H. SENSATION	/12
J. PASSIVE JOINT MOTION	/24
J. JOINT PAIN	/24

Action Research Arm Test

Patient Name:

Date:

Rater's Name:

Score Legend:

3	Finished the task quickly with the appropriate body position and normal hand/arm movements
2	Can finish the task but takes abnormally long, has great difficulty, or has improper body posture/hand/arm movements
1	Can finish the task partially regardless of position/movements
0	Unable to finish the task

Grasp

		Left	Right
1	Block (wood), 10 cm. cube		
2	Block (wood), 2.5 cm. cube		
3	Block (wood), 5 cm. cube		
4	Block (wood), 7.5 cm. cube		
5	Cricket ball		
6	Sharpening stone		
	Subtotal		

Grip

		Left	Right
1	Pour water from one glass to another		
2	Transfer a 2.25 cm tube from one side to the other		
3	Transfer a 1 cm. Tube from one side to the other		
4	Place the washer over the bolt		

		Left	Right
1	Ball Bearing between thumb and ring finger (3rd finger)		
2	Marble between thumb and index finger (1st finger)		
3	Ball bearing between thumb and middle finger (2nd finger)		
4	Ball bearing between thumb and index finger (1st finger)		
5	Marble between the thumb and ring finger (3rd finger)		
6	Marble between the thumb and ring finger (2nd finger)		
	Subtotal		

Gross Movement

		Left	Right
1	Hand behind the head		
2	Hand on top of the head		
3	Hand to mouth		
	Subtotal		

Total		
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Notes

Box and Blocks Testing Form

Name: _____

Dominant Hand (circle one): Right Left

Number of blocks transported in one minute:

Date: _____ Dominant Hand: _____ Non-Dominant Hand: _____

Date: _____ Dominant Hand: _____ Non-Dominant Hand: _____

Date: _____ Dominant Hand: _____ Non-Dominant Hand: _____

Date: _____ Dominant Hand: _____ Non-Dominant Hand: _____

Chedoke Arm and Hand Activity Inventory: Score Form
CAHAI-7 Version

Name: Date:

Activity Scale			
1. total assist (weak U/L < 25%) 2. maximal assist (weak U/L = 25-49%) 3. moderate assist (weak U/L = 50-74%) 4. minimal assist (weak U/L > 75%) 5. supervision 6. modified independence (device) 7. complete independence (timely, safely)			
Affected Limb:			Score
1. Open jar of coffee	☐ holds jar	☐ holds lid	
2. Call 911	☐ holds receiver	☐ dials phone	
3. Draw a line with a ruler	☐ holds ruler	☐ holds pen	
4. Pour a glass of water	☐ holds glass	☐ holds pitcher	
5. Wring out washcloth			
6. Do up five buttons			
7. Dry back with towel	☐ reaches for towel	☐ grasps towel end	
Total Score			/49
Comments			

COPY FREELY - DO NOT CHANGE
Copyright 2004 Chedoke Arm and Hand Activity Inventory, Hamilton, ON
Funded by The Ontario Ministry of Health and Long Term Care

Modified Ashworth Scale Instructions

General Information (derived Bohannon and Smith, 1987):

- Place the patient in a supine position
- If testing a muscle that primarily flexes a joint, place the joint in a maximally flexed position and move to a position of maximal extension over one second (count "one thousand one")
- If testing a muscle that primarily extends a joint, place the joint in a maximally extended position and move to a position of maximal flexion over one second (count "one thousand one")
- Score based on the classification below

Scoring (taken from Bohannon and Smith, 1987):

- | | |
|----|---|
| 0 | No increase in muscle tone |
| 1 | Slight increase in muscle tone, manifested by a catch and release or by minimal resistance at the end of the range of motion when the affected part(s) is moved in flexion or extension |
| 1+ | Slight increase in muscle tone, manifested by a catch, followed by minimal resistance throughout the remainder (less than half) of the ROM |
| 2 | More marked increase in muscle tone through most of the ROM, but affected part(s) easily moved |
| 3 | Considerable increase in muscle tone, passive movement difficult |
| 4 | Affected part(s) rigid in flexion or extension |

Patient Instructions:

The patient should be instructed to relax.

Downloaded from www.rehabmeasures.org

Test instructions provided courtesy of Richard Bohannon PT, PhD and Melissa Smith, PT
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Modified Ashworth Scale Testing Form

Name: _____ Date: _____

<u>Muscle Tested</u>	<u>Score</u>
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_____	_____
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_____	_____
-------	-------

_____	_____
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_____	_____
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_____	_____
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_____	_____
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_____	_____
-------	-------

_____	_____
-------	-------

_____	_____
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UAB Training for CI Therapy

SID _____ Name _____ Date _____ Visit _____ Examiner _____

Motor Activity Log (UE MAL) Score Sheet**Amount Scale How Well Scale**

- | | | | |
|---|-------|-------|--|
| 1. Turn on a light with a light switch | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 2. Open drawer | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 3. Remove an item of clothing from a drawer | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 4. Pick up phone | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 5. Wipe off a kitchen counter or other surface | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 6. Get out of a car
<i>(includes only the movement needed to get body from sitting to standing outside of the car, once the door is open).</i> | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 7. Open refrigerator | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 8. Open a door by turning a door knob/handle | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 9. Use a TV remote control | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 10. Wash your hands
<i>(includes lathering and rinsing hands; does not include turning water on and off with a faucet handle).</i> | _____ | _____ | if no, why? (use code) _____
Comments _____ |

Codes for recording “no” responses:

1. “I used the unaffected arm entirely.” (assign “0”).
2. “Someone else did it for me.” (assign “0”).
3. “I never do that activity, with or without help from someone else because it is impossible.” For example, combing hair for people who are bald. (assign “N/A” and drop from list of items).
4. “I sometimes do that activity, but did not have the opportunity since the last time I answered these questions.” (carry-over last assigned number for that activity).
5. Non-dominant hand hemiparesis. (only applicable to #24; assign “N/A” and drop from list of items).

UAB Training for CI Therapy

SID _____ Name _____ Date _____ Visit _____ Examiner _____

Amount Scale How Well Scale

- | | | | |
|---|-------|-------|--|
| 11. Turning water on/off
with knob/lever on faucet | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 12. Dry your hands | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 13. Put on your socks | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 14. Take off your socks | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 15. Put on your shoes
<i>(includes tying shoestrings and fastening straps)</i> | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 16. Take off your shoes
<i>(includes untying shoestrings and unfastening straps)</i> | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 17. Get up from a chair
with armrests | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 18. Pull chair away from
table before sitting down | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 19. Pull chair toward table
after sitting down | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 20. Pick up a glass, bottle,
drinking cup, or can <i>(does not need
to include drinking)</i> | _____ | _____ | if no, why? (use code) _____
Comments _____ |

Codes for recording “no” responses:

1. “I used the unaffected arm entirely.” (assign “0”).
2. “Someone else did it for me.” (assign “0”).
3. “I never do that activity, with or without help from someone else because it is impossible.” For example, combing hair for people who are bald. (assign “N/A” and drop from list of items).
4. “I sometimes do that activity, but did not have the opportunity since the last time I answered these questions.” (carry-over last assigned number for that activity).
5. Non-dominant hand hemiparesis. (only applicable to #24; assign “N/A” and drop from list of items).

UAB Training for CI Therapy

SID _____ Name _____ Date _____ Visit _____ Examiner _____

Amount Scale How Well Scale

- | | | | |
|---|-------|-------|--|
| 21. Brush your teeth
<i>(does not include preparation of toothbrush or brushing dentures unless the dentures are brushed while left in the mouth)</i> | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 22. Put on makeup base, lotion, or shaving cream on face | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 23. Use a key to unlock a door | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 24. Write on paper
<i>(If hand used to write pre-stroke is more affected, score item; if non-writing hand pre-stroke is more affected, drop item and assign N/A)</i> | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 25. Carry an object in your hand <i>(draping an item over the arm is not acceptable)</i> | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 26. Use a fork or spoon for eating <i>(refers to the action of bringing food to the mouth with fork or spoon)</i> | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 27. Comb your hair | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 28. Pick up a cup by a handle | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 29. Button a shirt | _____ | _____ | if no, why? (use code) _____
Comments _____ |
| 30. Eat half a sandwich or finger foods | _____ | _____ | if no, why? (use code) _____
Comments _____ |

Codes for recording “no” responses:

1. “I used the unaffected arm entirely.” (assign “0”).
2. “Someone else did it for me.” (assign “0”).
3. “I never do that activity, with or without help from someone else because it is impossible.” For example, combing hair for people who are bald. (assign “N/A” and drop from list of items).
4. “I sometimes do that activity, but did not have the opportunity since the last time I answered these questions.” (carry-over last assigned number for that activity).
5. Non-dominant hand hemiparesis. (only applicable to #24; assign “N/A” and drop from list of items).

Amount Scale (AS)

0 - Did not use my weaker arm (**not used**).

.5

1 - Occasionally used my weaker arm, but only very rarely (**very rarely**).

1.5

2 - Sometimes used my weaker arm but did the activity **most of the time** with my stronger arm (**rarely**).

2.5

3 - Used my weaker arm about half as much as before the stroke (**half pre-stroke**).

3.5

4 - Used my weaker arm almost as much as before the stroke (**3/4 pre-stroke**).

4.5

5 - Used my weaker arm as often as before the stroke (**same as pre-stroke**).

How Well Scale (HW)

0 - The weaker arm was not used at all for that activity
(never).

.5

1 - The weaker arm was moved during that activity
but was not helpful **(very poor).**

1.5

2 - The weaker arm was of some use during that
activity but needed some help from the stronger
arm or moved very slowly or with difficulty
(poor).

2.5

3 - The weaker arm was used for the purpose
indicated but movements were slow or were made
with only some effort **(fair).**

3.5

4 - The movements made by the weaker arm were
almost normal, but were not quite as fast or
accurate as normal **(almost normal).**

4.5

5 - The ability to use the weaker arm for that
activity was as good as before the stroke
(normal).

UAB Training for CI Therapy

Possible Reasons for Not Using the Weaker Arm for the Activity:

Reason A. “I used the unaffected arm entirely.”

Reason B. “Someone else did it for me.”.

Reason C. “I never do that activity, with or without help from someone else because it is impossible.” For example, combing hair for people who are bald.

Reason D. “I sometimes do that activity, but did not have the opportunity since the last time I answered these questions.”

Reason E. "That is an activity that I normally did only with my dominant hand before the stroke, and continue to do with my dominant hand now."

UAB Training for CI Therapy

Additional Items for the MAL-45

- Removing bills from a wallet
- Taking individual coins out of a pocket or purse
- Removing keys out of a pocket or purse
- Using a zipper pull
- Pouring liquid from a bottle
- Buckling a belt
- Popping top of beverage can
- Removing top from a medicine bottle
- Keypad press
- Use of keyboard/computer
- Putting on or taking off watch band
- Putting on glasses
- Pumping a soap dispenser
- Swiping a credit card or a card for an ATM
- Adjusting a home or hotel air conditioner or heater



Health Questionnaire

English version for India

Under each heading, please tick the ONE box that best describes your health TODAY.

MOBILITY

- I have no problems in walking ☐
- I have slight problems in walking ☐
- I have moderate problems in walking ☐
- I have severe problems in walking ☐
- I am unable to walk ☐

SELF-CARE

- I have no problems bathing or dressing myself ☐
- I have slight problems bathing or dressing myself ☐
- I have moderate problems bathing or dressing myself ☐
- I have severe problems bathing or dressing myself ☐
- I am unable to bathe or dress myself ☐

USUAL ACTIVITIES *(e.g. work, study, housework, family or leisure activities)*

- I have no problems doing my usual activities ☐
- I have slight problems doing my usual activities ☐
- I have moderate problems doing my usual activities ☐
- I have severe problems doing my usual activities ☐
- I am unable to do my usual activities ☐

PAIN / DISCOMFORT

- I have no pain or discomfort ☐
- I have slight pain or discomfort ☐
- I have moderate pain or discomfort ☐
- I have severe pain or discomfort ☐
- I have extreme pain or discomfort ☐

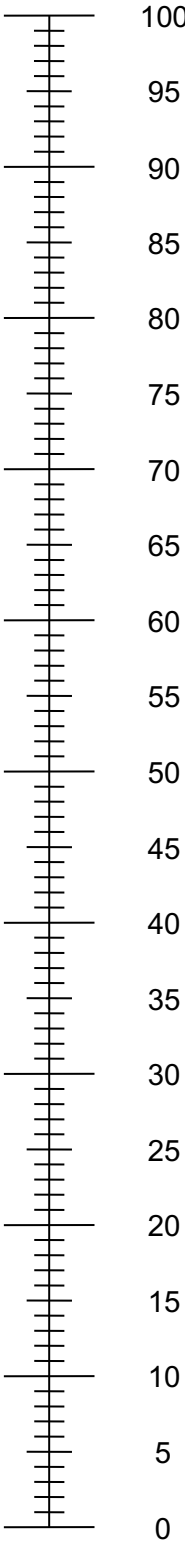
ANXIETY / DEPRESSION

- I am not anxious or depressed ☐
- I am slightly anxious or depressed ☐
- I am moderately anxious or depressed ☐
- I am severely anxious or depressed ☐
- I am extremely anxious or depressed ☐

- We would like to know how good or bad your health is TODAY.
- This scale is numbered from 0 to 100.
- 100 means the best health you can imagine.
0 means the worst health you can imagine.
- Mark an X on the scale to indicate how your health is TODAY.
- Now, please write the number you marked on the scale in the box below.

YOUR HEALTH TODAY =

The best health
you can imagine



The worst health
you can imagine

**University of North Carolina Hospitals
Chapel Hill, NC 27514
Department of Neurology**

**MODIFIED RANKIN SCALE (MRS)
MIM # 721**

Rater Name: _____

Date: _____

Score Description

- | | |
|---|---|
| 0 | No symptoms at all |
| 1 | No significant disability despite symptoms; able to carry out all usual duties and activities |
| 2 | Slight disability; unable to carry out all previous activities, but able to look after own affairs without assistance |
| 3 | Moderate disability; requiring some help, but able to walk without assistance |
| 4 | Moderately severe disability; unable to walk without assistance and unable to attend to own bodily needs without assistance |
| 5 | Severe disability; bedridden, incontinent and requiring constant nursing care and attention |
| 6 | Dead |

TOTAL (0–6): _____

Rater Sign / pager: _____

Provider Sign / MD# / pager: _____

HDF 536 / 4/03

References

Rankin J. "Cerebral vascular accidents in patients over the age of 60." *Scott Med J* 1957;2:200-15
Bonita R, Beaglehole R. "Modification of Rankin Scale: Recovery of motor function after stroke." *Stroke* 1988 Dec;19(12):1497-1500
Van Swieten JC, Koudstaal PJ, Visser MC, Schouten HJ, van Gijn J. "Interobserver agreement for the assessment of handicap in stroke patients." *Stroke* 1988;19(5):604-7

Caregiver Strain Index (CSI)

I am going to read a list of things that other people have found to be difficult. **Would you tell me whether any of these apply to you?** (GIVE EXAMPLES)

	Yes = 1	No = 0
Sleep is disturbed (e.g., because . . . is in and out of bed or wanders around at night)		
It is inconvenient (e.g., because helping takes so much time or it's a long drive over to help)		
It is a physical strain (e.g., because of lifting in and out of a chair; effort or concentration is required)		
It is confining (e.g., helping restricts free time or cannot go visiting)		
There have been family adjustments (e.g., because helping has disrupted routine; there has been no privacy)		
There have been changes in personal plans (e.g., had to turn down a job; could not go on vacation)		
There have been emotional adjustments (e.g., because of severe arguments)		
Some behavior is upsetting (e.g., because of incontinence; . . . has trouble remembering things; or . . . accuses people of taking things)		
It is upsetting to find . . . has changed so much from his/her former self (e.g., he/she is a different person than he/she used to be)		
There have been work adjustments (e.g., because of having to take time off)		
It is a financial strain		
Feeling completely overwhelmed (e.g., because of worry about . . . ; concerns about how you will manage)		
Total Score (Count yes responses. Any positive answer may indicate a need for intervention in that area. A score of 7 or higher indicates a high level of stress.)		

Robinson, B. (1983). Validation of a Caregiver Strain Index. *Journal of Gerontology*. 38:344-348.
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Appendix 2 – Subjective Index of Physical and Social Outcome

Please answer all questions

1. Since your stroke, how much difficulty do you have dressing yourself fully?

(Circle One Number)

No difficulty at all4
 Slight difficulty3
 Some difficulty2
 A lot of difficulty1
 I cannot dress myself fully0

2. Since your stroke, how much difficulty do you have moving around *all* areas of the home?

(Circle One Number)

No difficulty at all4
 Slight difficulty3
 Some difficulty2
 A lot of difficulty1
 I cannot move around all areas of the home0

3. Since your stroke, how satisfied are you with your overall ability to perform daily activities *in and around the home*?

(Circle One Number)

Completely satisfied4
 Mostly satisfied3
 Fairly satisfied2
 Not very satisfied1
 Completely dissatisfied0

4. Since your stroke, how much difficulty do you have shopping for and carrying *a few items* (1 bag of shopping or less) when at the shops?

(Circle One Number)

No difficulty at all4
 Slight difficulty3
 Some difficulty2
 A lot of difficulty1
 I cannot shop for and carry a few items0

5. Since your stroke, how independent are you in your ability to *move around your local neighbourhood*?

(Circle One Number)

I am completely independent4
 I prefer to have someone else with me3
 I need occasional assistance from someone2
 I need assistance much of the time1
 I am completely dependent on others0

6. Since your stroke, how often do you feel bored with your free time at home?

(Circle One Number)

I am never bored with my free time4
 A little of my free time3
 Some of my free time2
 Most of my free time1
 All of my free time0

7. Since your stroke, how would you describe the amount of communication between you and your friends/associates?

(Circle One Number)

A great deal4
 Quite a lot3
 Some2
 A little bit1
 None0

8. Since your stroke, how satisfied are you with the level of interests and activities you share with your friends/associates?

(Circle One Number)

Completely satisfied4
 Mostly satisfied3
 Fairly satisfied2
 Not very satisfied1
 Completely dissatisfied0

9. Since your stroke, how often do *you* visit friends/others?

(Circle One Number)

Most days4
 At least once a week3
 At least once a fortnight2
 Once a month or less1
 Never0

10. Since your stroke, how do you feel about your appearance when out in public?

(Circle One Number)

Perfectly happy4
 Slightly self-conscious3
 Fairly self-conscious2
 Very self-conscious1
 I try to avoid going out in public0

Thank you for completing this questionnaire.

PATIENT HEALTH QUESTIONNAIRE-9
(PHQ-9)

Over the last 2 weeks, how often have you had any of the following problems?
(Use "✓" to indicate your answer)

	Not at all	Several days	More than half the number of days	Almost every day
1. Little interest or pleasure in doing things	0	1	2	3
2. Feeling down, depressed, or hopeless	0	1	2	3
3. Trouble falling or staying asleep, or sleeping too much	0	1	2	3
4. Feeling tired or having little energy	0	1	2	3
5. Poor appetite or overeating	0	1	2	3
6. Feeling bad about yourself — or that you are a failure or have let yourself down or your family down	0	1	2	3
7. Trouble concentrating on things, such as reading the newspaper or watching television	0	1	2	3
8. Moving or speaking so slowly that other people could have noticed? Or the opposite — being so fidgety or restless that you have been moving around a lot more than usual	0	1	2	3
9. Thoughts that you would be better off dead or of hurting yourself in some way	0	1	2	3

FOR OFFICE CODING 0 + + +
=Total Score:

If you ticked off any problems, how difficult have these problems made it for you to do your work, take care of things at home, or get along with other people?

Not difficult at all	Somewhat difficult	Very difficult	Extremely difficult
☐	☐	☐	☐

NIH STROKE SCALE

Patient Identification. ____ - ____ - ____

Pt. Date of Birth ____ / ____ / ____

Hospital ____ (____ - ____)

Date of Exam ____ / ____ / ____

Interval: ☐ Baseline ☐ 2 hours post treatment ☐ 24 hours post onset of symptoms ± 20 minutes ☐ 7-10 days
☐ 3 months ☐ Other ____ (____)

Time: ____ : ____ [] am [] pm

Person Administering Scale _____

Administer stroke scale items in the order listed. Record performance in each category after each subscale exam. Do not go back and change scores. Follow directions provided for each exam technique. Scores should reflect what the patient does, not what the clinician thinks the patient can do. The clinician should record answers while administering the exam and work quickly. Except where indicated, the patient should not be coached (i.e., repeated requests to patient to make a special effort).

Instructions	Scale Definition	Score
1a. Level of Consciousness: The investigator must choose a response if a full evaluation is prevented by such obstacles as an endotracheal tube, language barrier, orotracheal trauma/bandages. A 3 is scored only if the patient makes no movement (other than reflexive posturing) in response to noxious stimulation.	0 = Alert; keenly responsive. 1 = Not alert; but arousable by minor stimulation to obey, answer, or respond. 2 = Not alert; requires repeated stimulation to attend, or is obtunded and requires strong or painful stimulation to make movements (not stereotyped). 3 = Responds only with reflex motor or autonomic effects or totally unresponsive, flaccid, and areflexic.	_____
1b. LOC Questions: The patient is asked the month and his/her age. The answer must be correct - there is no partial credit for being close. Aphasic and stuporous patients who do not comprehend the questions will score 2. Patients unable to speak because of endotracheal intubation, orotracheal trauma, severe dysarthria from any cause, language barrier, or any other problem not secondary to aphasia are given a 1. It is important that only the initial answer be graded and that the examiner not "help" the patient with verbal or non-verbal cues.	0 = Answers both questions correctly. 1 = Answers one question correctly. 2 = Answers neither question correctly.	_____
1c. LOC Commands: The patient is asked to open and close the eyes and then to grip and release the non-paretic hand. Substitute another one step command if the hands cannot be used. Credit is given if an unequivocal attempt is made but not completed due to weakness. If the patient does not respond to command, the task should be demonstrated to him or her (pantomime), and the result scored (i.e., follows none, one or two commands). Patients with trauma, amputation, or other physical impediments should be given suitable one-step commands. Only the first attempt is scored.	0 = Performs both tasks correctly. 1 = Performs one task correctly. 2 = Performs neither task correctly.	_____
2. Best Gaze: Only horizontal eye movements will be tested. Voluntary or reflexive (oculocephalic) eye movements will be scored, but caloric testing is not done. If the patient has a conjugate deviation of the eyes that can be overcome by voluntary or reflexive activity, the score will be 1. If a patient has an isolated peripheral nerve paresis (CN III, IV or VI), score a 1. Gaze is testable in all aphasic patients. Patients with ocular trauma, bandages, pre-existing blindness, or other disorder of visual acuity or fields should be tested with reflexive movements, and a choice made by the investigator. Establishing eye contact and then moving about the patient from side to side will occasionally clarify the presence of a partial gaze palsy.	0 = Normal. 1 = Partial gaze palsy; gaze is abnormal in one or both eyes, but forced deviation or total gaze paresis is not present. 2 = Forced deviation, or total gaze paresis not overcome by the oculocephalic maneuver.	_____

NIH STROKE SCALE

Patient Identification. ____ - ____ - ____

Pt. Date of Birth ____ / ____ / ____

Hospital ____ (____ - ____)

Date of Exam ____ / ____ / ____

Interval: ☐ Baseline ☐ 2 hours post treatment ☐ 24 hours post onset of symptoms $\pm$ 20 minutes ☐ 7-10 days
☐ 3 months ☐ Other ____ (____)

3. Visual: Visual fields (upper and lower quadrants) are tested by confrontation, using finger counting or visual threat, as appropriate. Patients may be encouraged, but if they look at the side of the moving fingers appropriately, this can be scored as normal. If there is unilateral blindness or enucleation, visual fields in the remaining eye are scored. Score 1 only if a clear-cut asymmetry, including quadrantanopia, is found. If patient is blind from any cause, score 3. Double simultaneous stimulation is performed at this point. If there is extinction, patient receives a 1, and the results are used to respond to item 11.	0 = No visual loss. 1 = Partial hemianopia. 2 = Complete hemianopia. 3 = Bilateral hemianopia (blind including cortical blindness).	_____
4. Facial Palsy: Ask – or use pantomime to encourage – the patient to show teeth or raise eyebrows and close eyes. Score symmetry of grimace in response to noxious stimuli in the poorly responsive or non-comprehending patient. If facial trauma/bandages, orotracheal tube, tape or other physical barriers obscure the face, these should be removed to the extent possible.	0 = Normal symmetrical movements. 1 = Minor paralysis (flattened nasolabial fold, asymmetry on smiling). 2 = Partial paralysis (total or near-total paralysis of lower face). 3 = Complete paralysis of one or both sides (absence of facial movement in the upper and lower face).	_____
5. Motor Arm: The limb is placed in the appropriate position: extend the arms (palms down) 90 degrees (if sitting) or 45 degrees (if supine). Drift is scored if the arm falls before 10 seconds. The aphasic patient is encouraged using urgency in the voice and pantomime, but not noxious stimulation. Each limb is tested in turn, beginning with the non-paretic arm. Only in the case of amputation or joint fusion at the shoulder, the examiner should record the score as untestable (UN), and clearly write the explanation for this choice.	0 = No drift; limb holds 90 (or 45) degrees for full 10 seconds. 1 = Drift; limb holds 90 (or 45) degrees, but drifts down before full 10 seconds; does not hit bed or other support. 2 = Some effort against gravity; limb cannot get to or maintain (if cued) 90 (or 45) degrees, drifts down to bed, but has some effort against gravity. 3 = No effort against gravity; limb falls. 4 = No movement. UN = Amputation or joint fusion, explain: _____ 5a. Left Arm 5b. Right Arm	_____ _____
6. Motor Leg: The limb is placed in the appropriate position: hold the leg at 30 degrees (always tested supine). Drift is scored if the leg falls before 5 seconds. The aphasic patient is encouraged using urgency in the voice and pantomime, but not noxious stimulation. Each limb is tested in turn, beginning with the non-paretic leg. Only in the case of amputation or joint fusion at the hip, the examiner should record the score as untestable (UN), and clearly write the explanation for this choice.	0 = No drift; leg holds 30-degree position for full 5 seconds. 1 = Drift; leg falls by the end of the 5-second period but does not hit bed. 2 = Some effort against gravity; leg falls to bed by 5 seconds, but has some effort against gravity. 3 = No effort against gravity; leg falls to bed immediately. 4 = No movement. UN = Amputation or joint fusion, explain: _____ 6a. Left Leg 6b. Right Leg	_____

NIH STROKE SCALE

Patient Identification. _____

Pt. Date of Birth ____/____/____

Hospital _____(____-____)

Date of Exam ____/____/____

Interval: ☐ Baseline ☐ 2 hours post treatment ☐ 24 hours post onset of symptoms $\pm$ 20 minutes ☐ 7-10 days
☐ 3 months ☐ Other _____(____)

7. Limb Ataxia: This item is aimed at finding evidence of a unilateral cerebellar lesion. Test with eyes open. In case of visual defect, ensure testing is done in intact visual field. The finger-nose-finger and heel-shin tests are performed on both sides, and ataxia is scored only if present out of proportion to weakness. Ataxia is absent in the patient who cannot understand or is paralyzed. Only in the case of amputation or joint fusion, the examiner should record the score as untestable (UN), and clearly write the explanation for this choice. In case of blindness, test by having the patient touch nose from extended arm position.	0 = Absent. 1 = Present in one limb. 2 = Present in two limbs. UN = Amputation or joint fusion, explain: _____	_____
8. Sensory: Sensation or grimace to pinprick when tested, or withdrawal from noxious stimulus in the obtunded or aphasic patient. Only sensory loss attributed to stroke is scored as abnormal and the examiner should test as many body areas (arms [not hands], legs, trunk, face) as needed to accurately check for hemisensory loss. A score of 2, "severe or total sensory loss," should only be given when a severe or total loss of sensation can be clearly demonstrated. Stuporous and aphasic patients will, therefore, probably score 1 or 0. The patient with brainstem stroke who has bilateral loss of sensation is scored 2. If the patient does not respond and is quadriplegic, score 2. Patients in a coma (item 1a=3) are automatically given a 2 on this item.	0 = Normal; no sensory loss. 1 = Mild-to-moderate sensory loss; patient feels pinprick is less sharp or is dull on the affected side; or there is a loss of superficial pain with pinprick, but patient is aware of being touched. 2 = Severe to total sensory loss; patient is not aware of being touched in the face, arm, and leg.	_____
9. Best Language: A great deal of information about comprehension will be obtained during the preceding sections of the examination. For this scale item, the patient is asked to describe what is happening in the attached picture, to name the items on the attached naming sheet and to read from the attached list of sentences. Comprehension is judged from responses here, as well as to all of the commands in the preceding general neurological exam. If visual loss interferes with the tests, ask the patient to identify objects placed in the hand, repeat, and produce speech. The intubated patient should be asked to write. The patient in a coma (item 1a=3) will automatically score 3 on this item. The examiner must choose a score for the patient with stupor or limited cooperation, but a score of 3 should be used only if the patient is mute and follows no one-step commands.	0 = No aphasia; normal. 1 = Mild-to-moderate aphasia; some obvious loss of fluency or facility of comprehension, without significant limitation on ideas expressed or form of expression. Reduction of speech and/or comprehension, however, makes conversation about provided materials difficult or impossible. For example, in conversation about provided materials, examiner can identify picture or naming card content from patient's response. 2 = Severe aphasia; all communication is through fragmentary expression; great need for inference, questioning, and guessing by the listener. Range of information that can be exchanged is limited; listener carries burden of communication. Examiner cannot identify materials provided from patient response. 3 = Mute, global aphasia; no usable speech or auditory comprehension.	_____
10. Dysarthria: If patient is thought to be normal, an adequate sample of speech must be obtained by asking patient to read or repeat words from the attached list. If the patient has severe aphasia, the clarity of articulation of spontaneous speech can be rated. Only if the patient is intubated or has other physical barriers to producing speech, the examiner should record the score as untestable (UN), and clearly write an explanation for this choice. Do not tell the patient why he or she is being tested.	0 = Normal. 1 = Mild-to-moderate dysarthria; patient slurs at least some words and, at worst, can be understood with some difficulty. 2 = Severe dysarthria; patient's speech is so slurred as to be unintelligible in the absence of or out of proportion to any dysphasia, or is mute/anarthric. UN = Intubated or other physical barrier, explain: _____	_____

NIH

STROKE

SCALE

Patient Identification. _____

Pt. Date of Birth ____/____/____

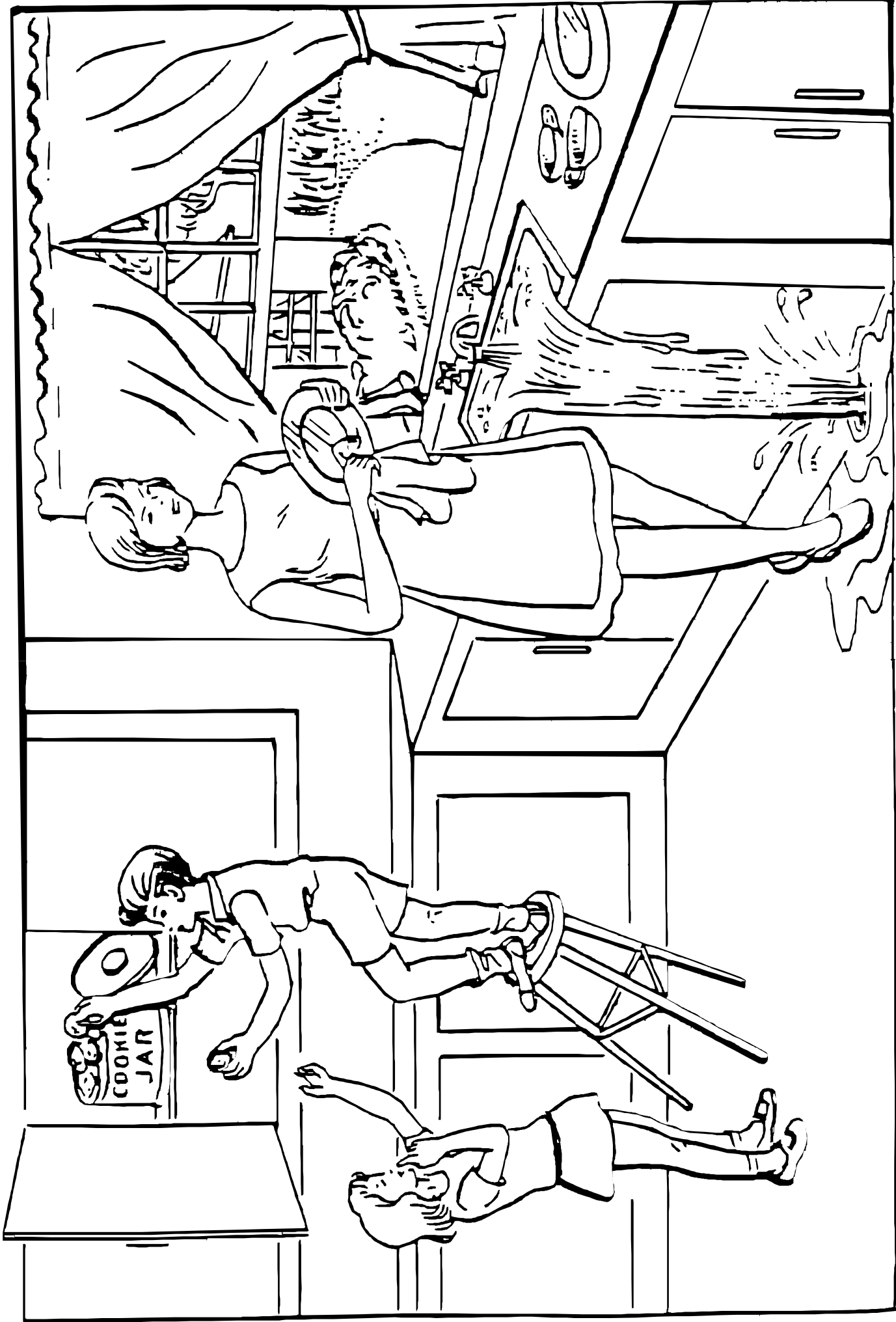
Hospital _____(____-____)

Date of Exam ____/____/____

Interval: ☐ Baseline ☐ 2 hours post treatment ☐ 24 hours post onset of symptoms ±20 minutes ☐ 7-10 days

☐ 3 months ☐ Other _____(____)

11. Extinction and Inattention (formerly Neglect): Sufficient information to identify neglect may be obtained during the prior testing. If the patient has a severe visual loss preventing visual double simultaneous stimulation, and the cutaneous stimuli are normal, the score is normal. If the patient has aphasia but does appear to attend to both sides, the score is normal. The presence of visual spatial neglect or anosagnosia may also be taken as evidence of abnormality. Since the abnormality is scored only if present, the item is never untestable.	0 = No abnormality. 1 = Visual, tactile, auditory, spatial, or personal inattention or extinction to bilateral simultaneous stimulation in one of the sensory modalities. 2 = Profound hemi-inattention or extinction to more than one modality; does not recognize own hand or orients to only one side of space.	_____ _____
---	--	-----------------------------------



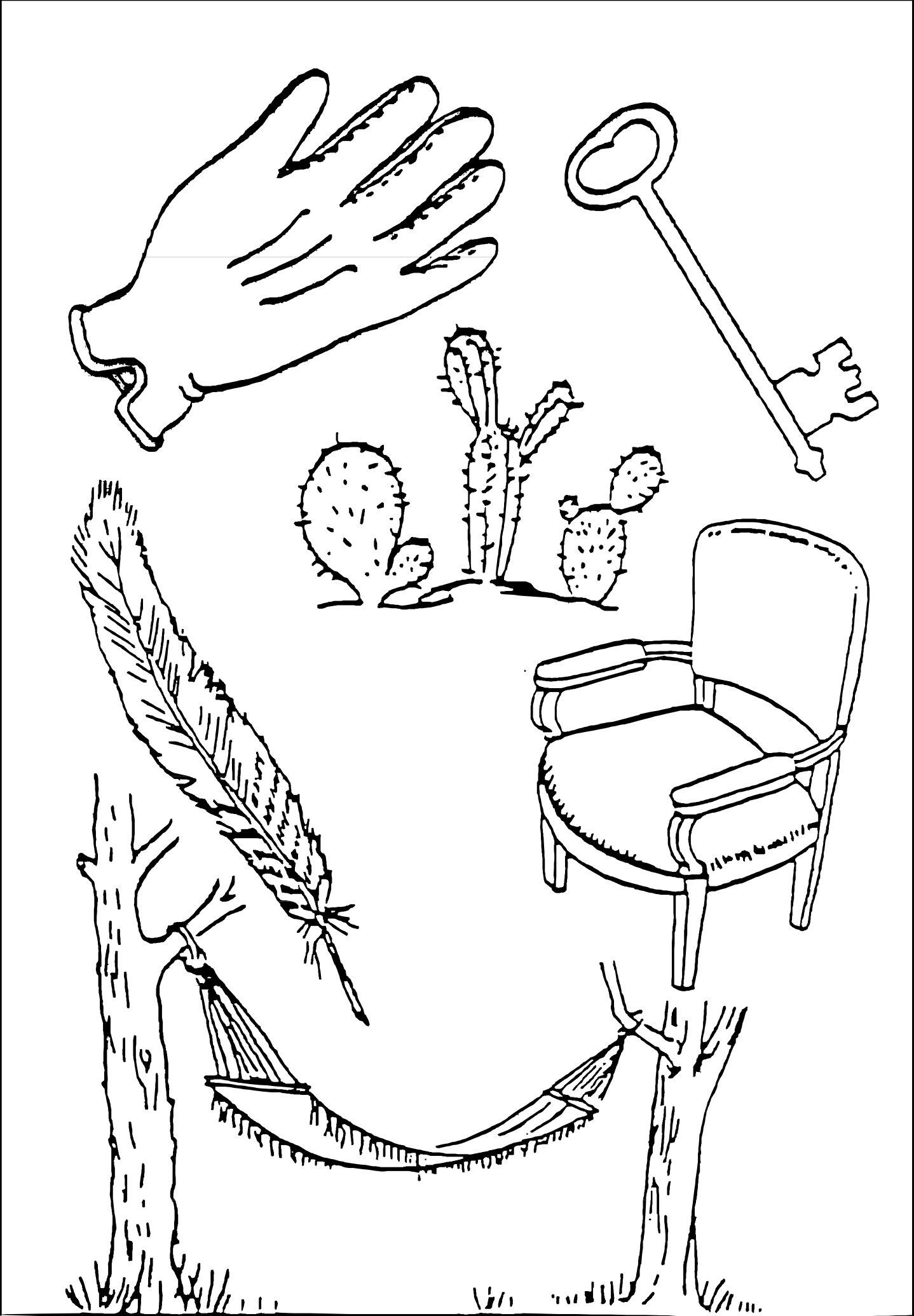
You know how.

Down to earth.

I got home from work.

**Near the table in the dining
room.**

**They heard him speak on the
radio last night.**



MAMA
TIP – TOP
FIFTY – FIFTY
THANKS
HUCKLEBERRY
BASEBALL PLAYER

Department of Bioengineering
Christian Medical College, Vellore

Informed Consent form to participate in a research study

Study Title: HOMER: Home-based robot-assisted upper-limb training for stroke in India.

Study Number: _____

Subject's Initials: _____

Subject's Name: _____

Date of Birth / Age: _____

- (i) I confirm that I have read and understood the information sheet dated _____ for the above study and have had the opportunity to ask questions. []
- (ii) I understand that my participation in the study is voluntary and that I am free to withdraw at any time, without giving any reason, without my medical care or legal rights being affected. []
- (iii) I understand that the Ethics Committee and the regulatory authorities will not need my permission to look at my health records both in respect of the current study and any further research that may be conducted in relation to it, even if I withdraw from the trial. I agree to this access. However, I understand that my identity will not be revealed in any information released to third parties or published. []
- (iv) I agree not to restrict the use of any data or results that arise from this study provided such a use is only for scientific purpose(s). []
- (v) I agree to take part in the above study. []
- (vi) I am aware of the Audio-visual recording of the Informed Consent. []

Signature (or Thumb impression) of the Subject/Legally Acceptable

Date: ____/____/____

Signatory's Name: _____

Signature:

Or

Representative: _____

Date: ____/____/____

Signatory's Name: _____

Signature of the Investigator: _____

Date: ____/____/____

Study Investigator's Name: _____

Signature or thumb impression of the Witness: _____

Date: ____/____/____

Name & Address of the Witness: _____

Department of Bioengineering
Christian Medical College, Vellore
Information sheet

Study Title: HOMER: Home-based robot-assisted upper-limb training for stroke in India

Date:

You are requested to participate in a study to evaluate if robot-assisted home-based arm and hand therapy is better than a standard home-based therapy program prescribed to patients with arm and hand impairments after a stroke. We have developed two robotic devices called PLUTO for assisted training of hand functions and MARS for assisted training of arm functions. We wish to test in this study if training with these devices leads to better recovery than a standard home therapy program that is used for arm and hand therapy. The result of this study can provide evidence for whether we need robots to provide arm and hand therapy at home.

If you take part, what will you have to do?

You will be randomly assigned to either undergo the standard home therapy program for four weeks or you will be provided a robot-based training program at home for four weeks. Your arm and hand movement ability will be assessed three times during the study.

You will undergo the following training and assessment as part of the study.

Training

Standard Home Therapy Program	Robot-Assisted Home Therapy Program
- You will be given training at home for three days by a therapist who will visit you at your home. - You will need to train for four weeks, 2 hours a day, 6 days a week by yourself without the direct supervision of therapy after the first three days. - You will have a weekly video call session with the therapist to review your progress.	- You will be given a PLUTO and MARS robot, which will be set up at your home for four weeks. - A therapist will supervise your therapy with the robots for the first three days at your home. - You will need to train for four weeks, 2 hours a day, 6 days a week by yourself without the direct supervision of therapy after the first three days. You will need to train with the robots for 1.5hrs and 0.5hrs of non-robotic therapy. - You will have a weekly video call session with the therapist to review your progress.

Assessments

- You will be assessed 3 times as part of this study as described below. Each assessment session will take a maximum of 60 minutes.
- **Assessment 0:** This is a baseline assessment carried out before the start of the training intervention. Here, the ability of your upper limb will be assessed using various standard clinical scales. The amount of movement you possess in your wrist and hand will be evaluated using PLUTO and MARS.

- **Assessment 1:** This assessment will be done immediately after completing the four-week training. In addition to assessing the ability of your upper limb, we will also assess your experience with the four-week training either with the robots or the home-therapy program. This assessment will tell us if your upper limb has improved from the training and your perception of the home therapy program with or without the robot.
- **Assessment 2:** This assessment will be done at the end of the 6th month from the day you had your stroke. The ability of your upper limb will be evaluated in this assessment. This will tell us if any gain in upper limb ability you made from the four-week training is maintained as you enter the chronic phase of your stroke.

If you are currently receiving any other therapy, you must continue that in addition to being part of the current study.

Does using the two robots have any side effects?

There are no side effects or harm from using these robots. The robots are safe to use and have all the necessary safety mechanisms to prevent the application of any unsafe forces to your hand. The robots also have emergency stop buttons under your control for completely shutting them down. We have already done short pilot usability studies where no adverse events were reported.

Can you withdraw from this study after it starts?

Your participation in this study is entirely voluntary, and you are free to withdraw your participation at any point. Your withdrawal from the study will not affect your usual treatment at this hospital if you are receiving one.

What will happen if you develop any study-related injury?

The training you will perform will not be very different from the normal activities that you perform during usual therapy. So, the chances of you developing study-related injury are very low.

Will you have to pay for the study session?

No, you will not be requested to pay for the study. You will be reimbursed for your travel to the hospital for carrying out Assessment-1 and Assessment-2.

What happens after the study is over?

The information collected from the study will be used to evaluate the effectiveness of a home-based robot-assisted training protocol with a hand and arm robot. This will allow us to identify the best way to incorporate robot-assisted therapy into home therapy.

Will your personal details be kept confidential?

The results of this study will be published in a medical journal, but you will not be identified by name in any publication or presentation of the results. Videos of your hand/arm during training will be recorded during therapy. These videos will not include your face. These videos might be used as part of publications and presentations of the study results. You can also opt out of this video recording.

If you have any further questions, please ask:

Dr. Sivakumar B, Phone: 9629826673;

Dr. Sanjith Aaron, Phone: 9894022395

Dr. Prabhakar AT, Phone: 9486197550